

Jaime López Ruiz

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[Portfolio: jaimelopezruiz.github.io](https://portfolio.jaimelopezruiz.github.io) | github.com/jaimelopezruiz | linkedin.com/in/jaimelopezruiz

EDUCATION

Imperial College London

Oct 2022 – Jun 2027

MEng Mechanical Engineering with a Year in Industry - Expected First Class Honours

- Relevant coursework: Robot Dynamics & Control, Machine Learning, Dynamics & Vibrations, Mechatronics and Control, Embedded C for Microcontrollers

PUBLICATIONS & PRESENTATIONS

Irradiation-induced damage in ion-irradiated SiC characterised by Raman spectroscopy

Manuscript in preparation for peer-reviewed submission, under supervision of Prof Dong Liu and Dr Alex Leide

High-temperature Raman spectroscopy of in-situ annealing of ion-irradiated SiC and residual stresses in SiC/SiC composites

Co-author, invited talk - Symposium on Spectroscopic Methods and Analysis for Nuclear Energy Related Materials, TMS 2027 Annual Meeting, Orlando

RESEARCH & PROJECTS

Markerless Robot Arm Teleoperation

Apr 2026 – Jun 2026

Personal project

- Built a markerless teleoperation pipeline mapping monocular MediaPipe pose estimation to real-time joint commands for a 5-DOF SO-101 arm; no wearables, IMUs or fiducial markers.
- Implemented Product-of-Exponentials kinematics for a 5-DOF arm: URDF-to-screw-axis extraction and a custom damped-least-squares IK with joint-limit clamping and null-space redundancy resolution, built around the Modern Robotics library's rigid-body/FK primitives.
- Validated numerically with condition-number analysis and convergence-rate benchmarking.

University of Oxford

Jun 2025 – Aug 2025

Undergraduate Researcher (UKAEA FOSTER Grant)

- Built a Python pipeline for quantitative Raman analysis of irradiated ceramics: baseline correction, config-driven region-wise fitting of overlapping peaks (Gaussian, Lorentzian, Voigt, Breit-Wigner-Fano), 1σ uncertainties propagated from the fit covariance and validated against Monte Carlo to 0.1%, and automated fit-quality diagnostics and flags.
- Designed uncertainty budgets for derived material metrics (chemical-disorder ratio, a coupling-invariant crystallinity index, and residual stress with a stated detection floor), separating statistical from systematic error; all reported values and figures regenerate from tracked code, configs and raw spectra.
- Investigated irradiation-induced damage in SiC under nuclear reactor conditions using Raman spectroscopy.

Autonomous Maze-Solving Buggy

2024 – 2025

Imperial College, team of 2

- Bare-metal C on PIC18: I2C colour-sensor driver with calibrated gain and integration timing, fixed-point RGB normalisation classifying 9 instruction cards.
- Stack-based path recording inverted to retrace the maze autonomously.

Reusable Liquid Bipropellant Rocket

Oct 2024 – Jun 2025

Lead Feed System & Structures Engineer

- First student rocket in UK history to hot fire vertically: led design, build and test of the propellant feed system and body structure for a 1 kN-thrust, 10 kN-s-impulse liquid bipropellant rocket.

INDUSTRY EXPERIENCE

PerkinElmer Ltd.

Sep 2025 – Aug 2026

Mechanical Design Engineer (12-month industrial placement) | Global scientific instrumentation manufacturer, High Wycombe, UK

- Designed a coupled, synchronised precision motor assembly with a pre-loaded friction-drive reduction, achieving 0.001° angular resolution in the instrument's optical path, and built the metrology to prove it, qualifying the axis against an autocollimator reference.
- Owned the motion-control subsystem end-to-end: Python/I²C driver and characterisation suite, step-scan synchronisation with an optical chopper (zero cumulative wavelength-axis drift), and the drive's optical acceptance criteria defined jointly with the optics team.
- Cut production test time by over 4 hours (30%) per instrument by designing and manufacturing custom test jigs and assembly accessories.
- Owned components from concept to delivery: rapid prototyping (3D printing, machined parts), procurement of standard, custom and motor components, applying GD&T, FMEA and poka-yoke.

SKILLS

Technical: Python, C, PyTorch, NumPy; robot kinematics and control; computer vision and pose estimation; embedded C, I2C; SolidWorks, GD&T, 3D printing

Languages: Spanish (native), English (native), French (fluent, DALF C1)